DoubleMLDeep: Estimation of Causal Effects with Multimodal Data

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Sven Klaassen
University of Hamburg
Economic AI

Jan Teichert-Kluge
University of Hamburg

Philipp Bach
University of Hamburg
Economic AI

Victor Chernozhukov
Massachusetts Institute of Technology

Martin Spindler
University of Hamburg
Economic AI

Suhas Vijaykumar
Massachusetts Institute of Technology

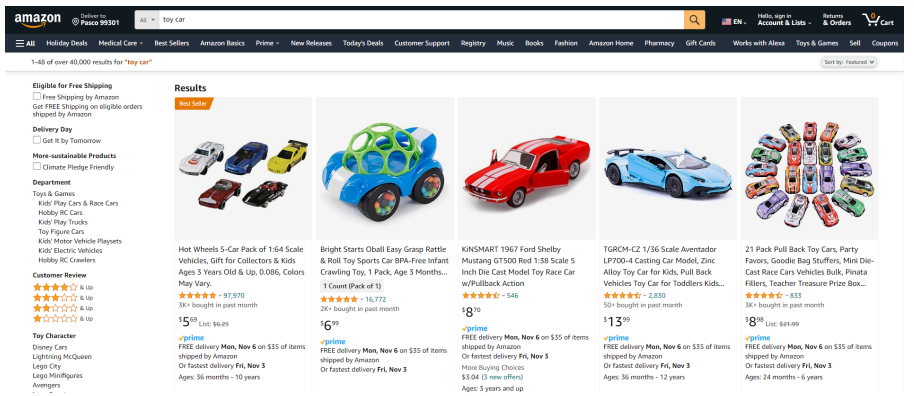
Motivation

Motivation

- Causal Inference mainly relies on tabular data

	A	B	C	D	E	F	G	H	I	J	K
1	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	Y
2	0,26222194	0,73438599	0,44667203	0,97138377	0,081064	0,28441456	0,30204598	0,42916989	0,45202869	0,48458485	0,69688241
3	0,02086748	0,76377164	0,82984213	0,44114913	0,6791843	0,10440879	0,06871173	0,65049866	0,38145381	0,37883627	0,68724919
4	0,80214042	0,60108851	0,3924631	0,64680441	0,52447716	0,71441139	0,92058036	0,50487157	0,0639613	0,72018174	0,70586344
5	0,69860016	0,48544746	0,02697306	0,56612716	0,64030572	0,38122836	0,77357736	0,7665354	0,56153968	0,40036439	0,96259876
6	0,76871281	0,52693951	0,23152259	0,99343286	0,19512117	0,52281687	0,74631891	0,51990814	0,8916446	0,13854693	0,15014547
7	0,80559982	0,49248903	0,52094313	0,83491804	0,19224862	0,51690107	0,99896405	0,21315395	0,13844972	0,5972433	0,30108879
8	0,14274622	0,68082916	0,21375762	0,12917741	0,101114	0,35962408	0,37101036	0,88481715	0,0917219	0,81144147	0,23748577
9	0,5565959	0,27805181	0,38191987	0,96746964	0,88262331	0,64204409	0,61391038	0,14798767	0,75991882	0,71364563	0,26831957
10	0,86680082	0,98940603	0,08032276	0,51284266	0,85067788	0,31879029	0,14212956	0,58889047	0,85003262	0,40984155	0,90320049
11	0,49294831	0,9930191	0,32376346	0,76486908	0,71150197	0,49307446	0,00732655	0,1366301	0,95413681	0,8192993	0,41659136
12	0,83367503	0,99203204	0,94500561	0,38442651	0,08267135	0,44319059	0,50597368	0,33789173	0,71150674	0,71723848	0,82498573
13	0,30901589	0,45442227	0,39182182	0,94058761	0,84662039	0,13237356	0,39203009	0,23863556	0,66904622	0,66739594	0,05703788
14	0,84854041	0,07220964	0,60564451	0,11235493	0,11423158	0,19425556	0,10525554	0,69400453	0,24273427	0,74521656	0,1934563
15	0,41204349	0,37890159	0,21172493	0,47214042	0,28972671	0,28045945	0,26144942	0,50182126	0,56848988	0,34171099	0,55046509
16	0,8270131	0,7550563	0,68080658	0,99718432	0,54748336	0,07330951	0,96389321	0,43492645	0,32444209	0,37929049	0,54724228
17	0,65968289	0,42809072	0,50651859	0,85136778	0,67725248	0,80776288	0,80154682	0,41961695	0,03794024	0,10550978	0,18250098
18	0,00250478	0,10976101	0,58665193	0,59539311	0,06932994	0,44532097	0,38697059	0,86107306	0,32445673	0,03897245	0,4593923
19	0,1646558	0,11563336	0,54131562	0,70212986	0,78935386	0,65759136	0,96083497	0,18585388	0,02245136	0,08388247	0,9684134

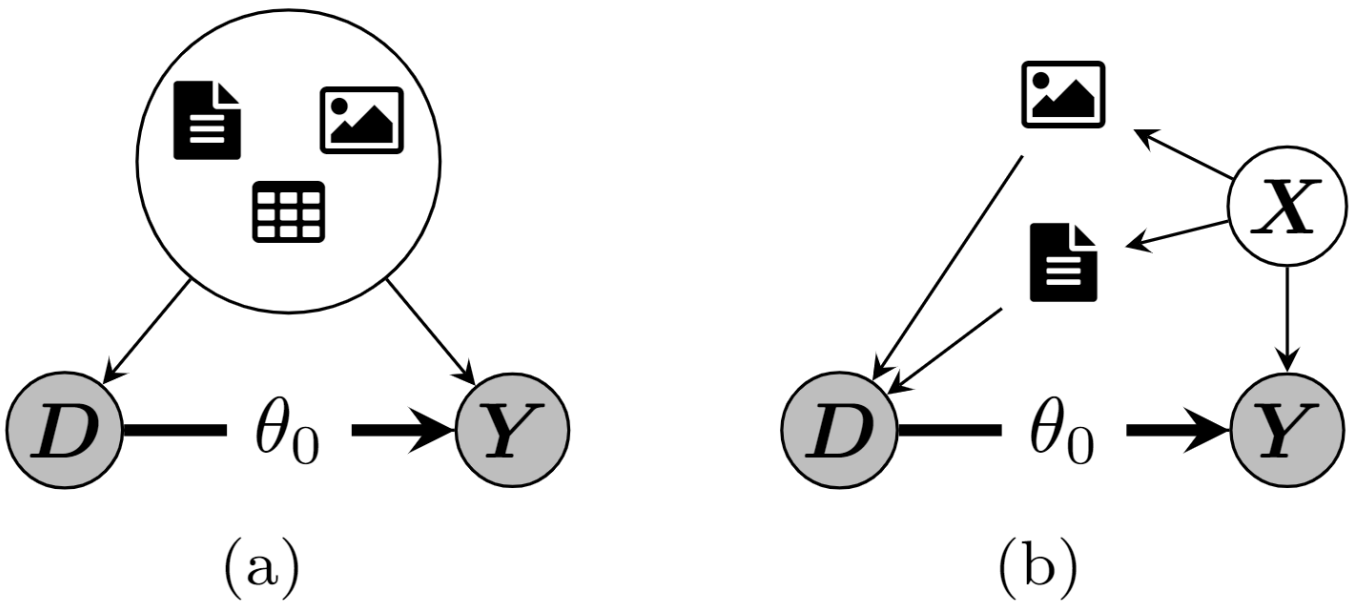
- In a lot of applications additional unstructured data is available



- We consider multimodal data as **confounders**
- Applications in Marketing, Medicine / Health, Finance, ….
  - Demand estimation
  - Estimation of treatment effects conditioning on X-ray images

Motivation

- Use multimodal data (text and images) *additionally* to conventional tabular features



Examples of directed acyclic graphs (DAGs) with image and text confounding. (a) Direct confounding via image, text and tabular data. (b) Treatment decision is driven by text and images. All backdoor paths are blocked by conditioning on both image and text data.

$\Rightarrow$ Adjust for confounding encoded in unstructured data

Motivation

Opportunities

- Exploit state-of-the art architectures for causal estimation (for example LLMs)
- Example: Demand estimation, account for description, images, reviews
- Huge potential for future research

Challenges

- Recent innovations in deep learning for prediction
- Limitations of ML-based estimation due to regularization in causal estimation
- Need for effective Deep Learning architecture
- Challenges in identifying causal effects even in idealized simulations

Our approach

- Integrate state-of-the art Deep Learning tools for estimation of Causal Parameters
- *DoubleML*[1]: Use DML to account for ML regularization bias
- Combination: *DoubleML* + Deep Learning = **DoubleMLDeep**

Literature Review

- Unstructured data like images and text are key for estimating causal parameters in various contexts, such as price elasticities and medical treatments (Chan et al. 2018; Masukawa et al. 2022), and climate effects (Wu, Duan, and Zhang 2023).
- Text as an outcome/treatment has been conceptually explored by Egami et al. (2018) and Sridhar and Blei (2022).
- Veitch, Sridhar, and Blei (2020) discuss consistency in this setting; we extend it by integrating text/images into double machine learning for valid inference.
- Transformer models for treatment effects are applied by Melnychuk, Frauen, and Feuerriegel (2022), though without inference results.
- There are also approaches to use images in causal inference (Jerzak, Johansson, and Daoud 2023a, 2023b, 2023c), but our approach enable valid inference with both, text and images, in a multimodal framework.

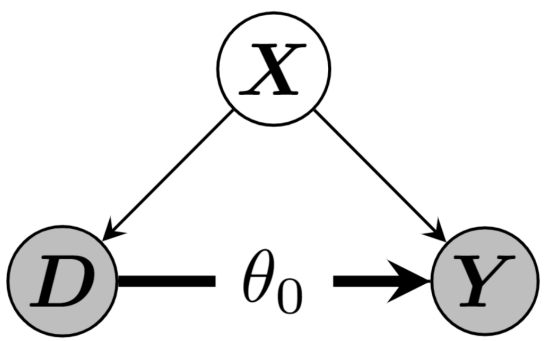
Motivating Example - PLR

Partially linear regression model (PLR)

$$
\begin{aligned}
Y &= \theta_0 D + g_0(X) + \varepsilon, & \mathbb{E}[\varepsilon | X, D] &= 0 \\
D &= m_0(X) + \vartheta, & \mathbb{E}[\vartheta | X] &= 0
\end{aligned}
$$

with

- Y - outcome variable
- D - policy/treatment variable
- $X = (X_1, \ldots X_p)^T$ - vector of additional controls
- ε, ϑ - stochastic errors



DAG for PLR Model

Double/Debiased Machine Learning (DML)

- Use machine learning methods to fit

$$l_0(X) := \mathbb{E}[Y|X]$$
$$m_0(X) := \mathbb{E}[D|X]$$

- Construct the orthogonalized score (either cross-fitting or split the sample) for a fitted nuisance learner $\hat{\eta} := (\hat{l}, \hat{m})$

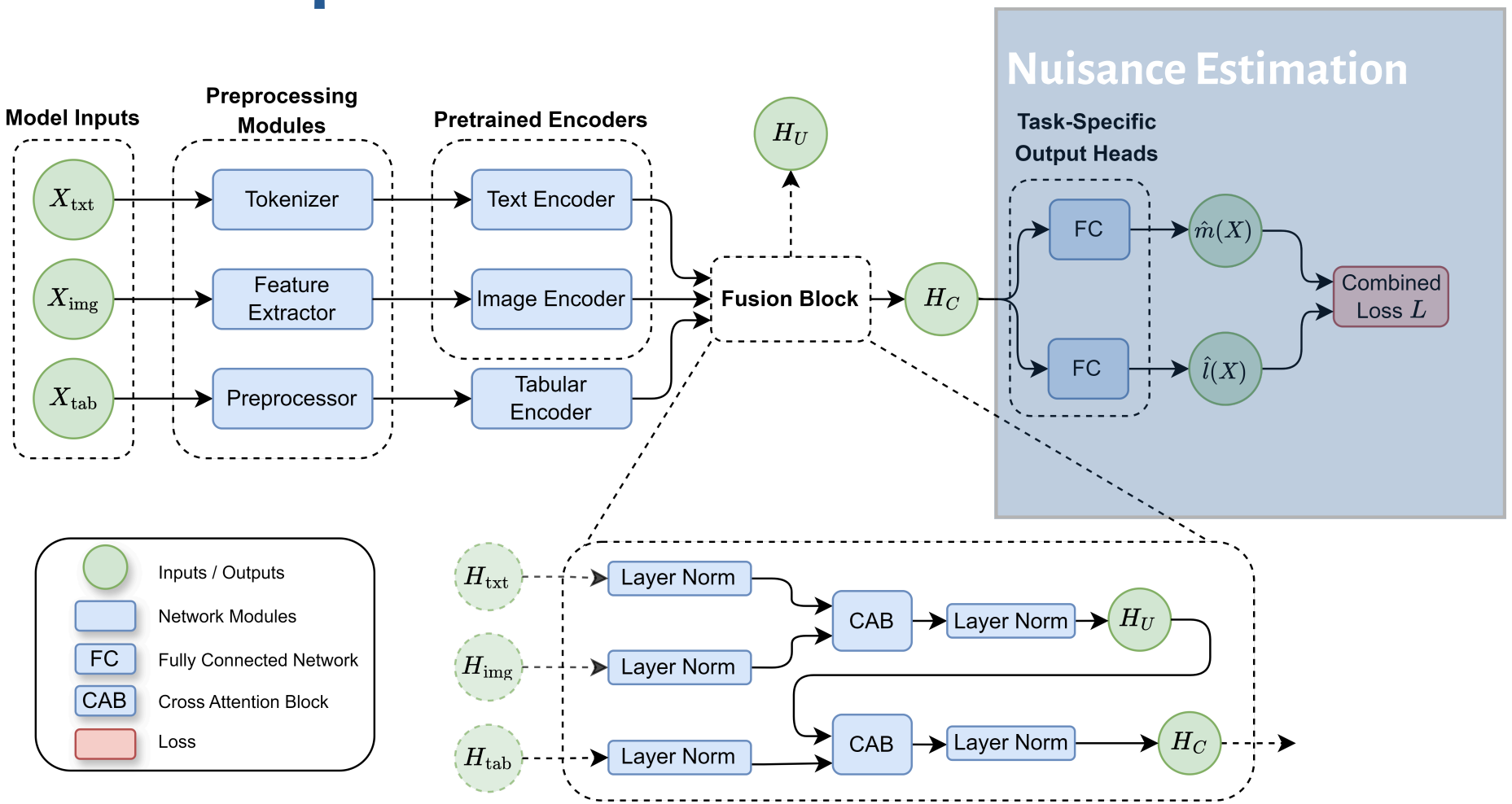
$$\psi(W, \theta, \hat{\eta}) := \Big(Y - \hat{l}(X) - \theta\big(D - \hat{m}(X)\big)\Big)\Big(D - \hat{m}(X)\Big)$$

- Compute the estimate as the solution of

$$0 = \frac{1}{n}\sum_{i=1}^{n} \psi(W, \hat{\theta}, \hat{\eta})$$

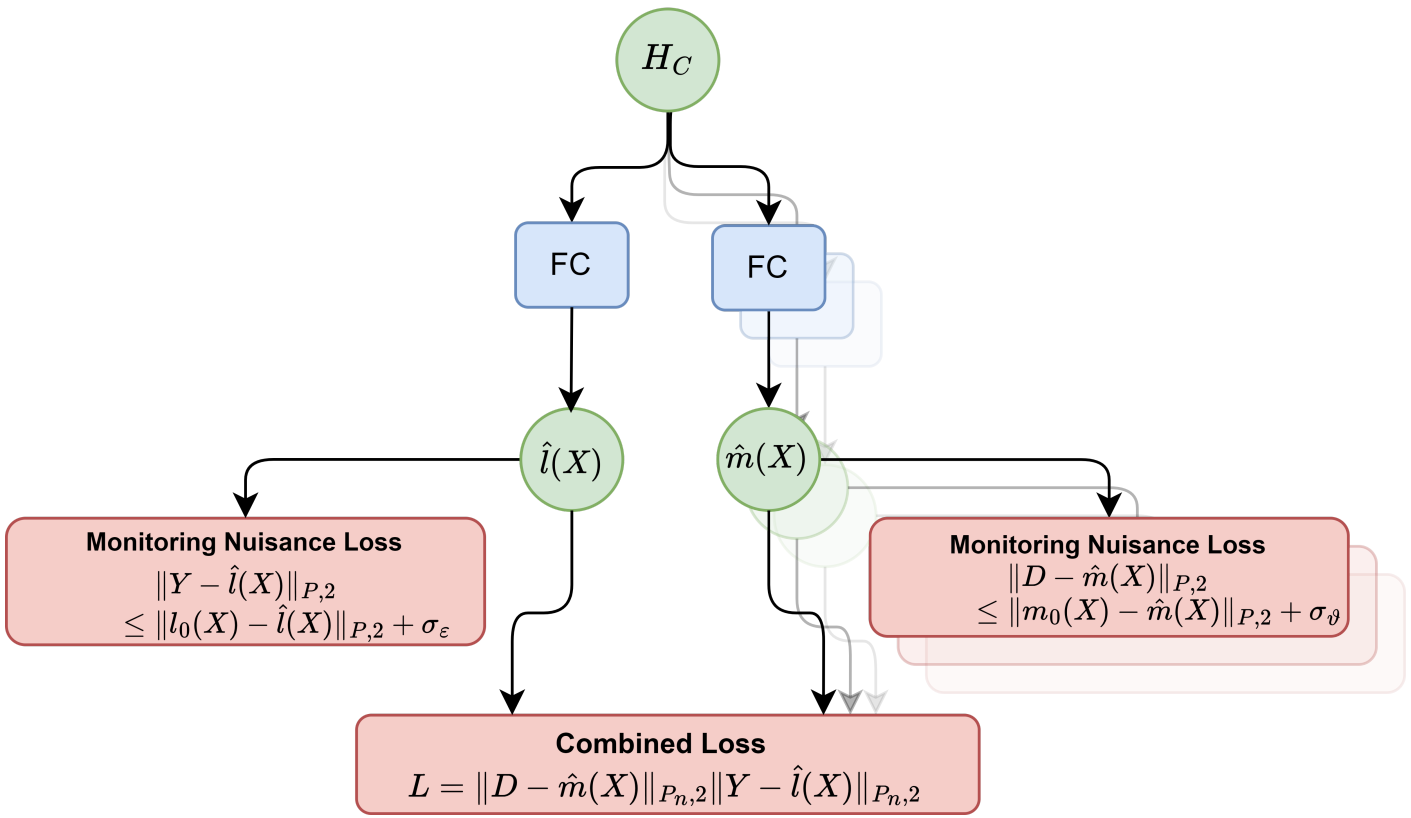
DoubleMLDeep

DoubleMLDeep - Module Structure



High-Level PLR Model Architecture. Both nuisance components are trained simultaneously with a combined loss.

DoubleMLDeep - Module Structure



Nuisance Estimation with Combined Loss in case of multiple treatments with nuisance loss monitoring.

- It is possible to use binary treatment variables $D \in \{0, 1\}$ by adjusting the loss function:

$$L = L_{log}(D, \hat{m}(X))\|Y - \hat{l}\,(X)\|_{P_n,2}$$

Application

Amazon Toys Dataset

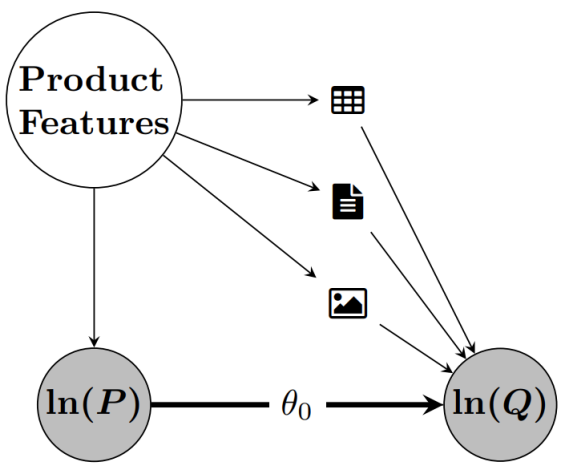
- Public Data from the Amazon sales platform from category toys and games for subcategories vehicles, cars and trucks.
- Using the Sales Rank as a proxy for quantities, as shown in Bajari et al. (2023).

Variable	Description
Sales Rank	Sales rank as weighted mean for the last 30 days
Price	Price as weighted mean for the last 30 days
Tabular	Tabular features as reviews, ratings, etc.
Text	Combination of Title, Category, Description, etc.
Image	Image of the product

Model

- Run a simple log-log regression model
- Images and Text block backdoor path from price to demand (or sales rank)

$$\ln(Q) = \theta_0 \ln(P) + g_0(X) + \epsilon$$



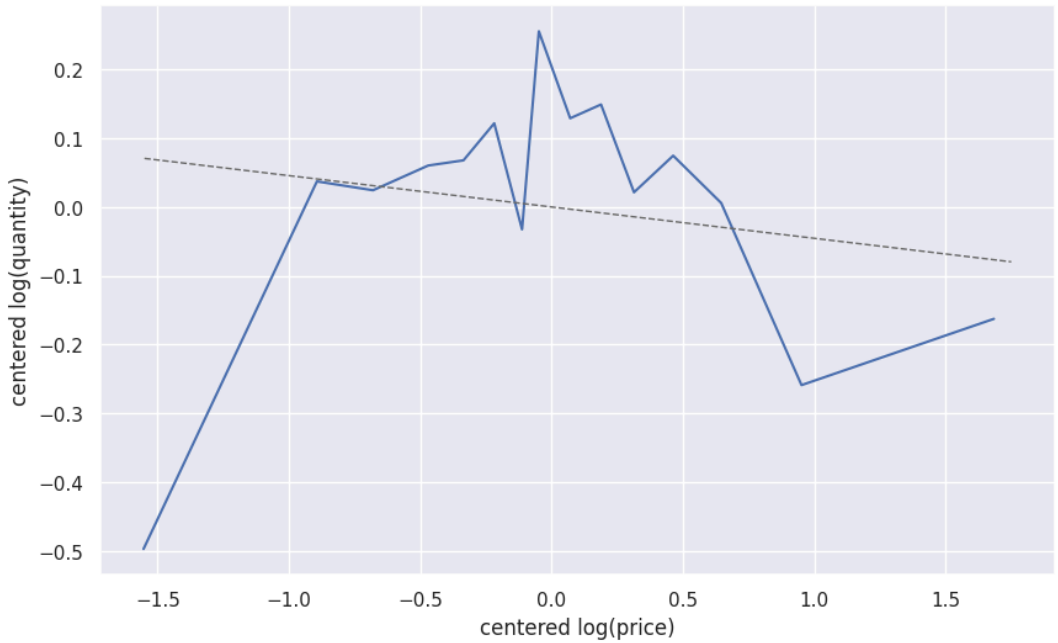
Baseline OLS Model

- Baseline estimate with tabular covariates X_{tab}
- OLS:

$$\ln(Q) = \theta_0 \ln(P) + \beta^T X_{\text{tab}} + \epsilon$$

- $R^2 = 0.330$

2.5 %	$\hat{\theta}$	**97.5 %**
-0.072	-0.046	-0.019

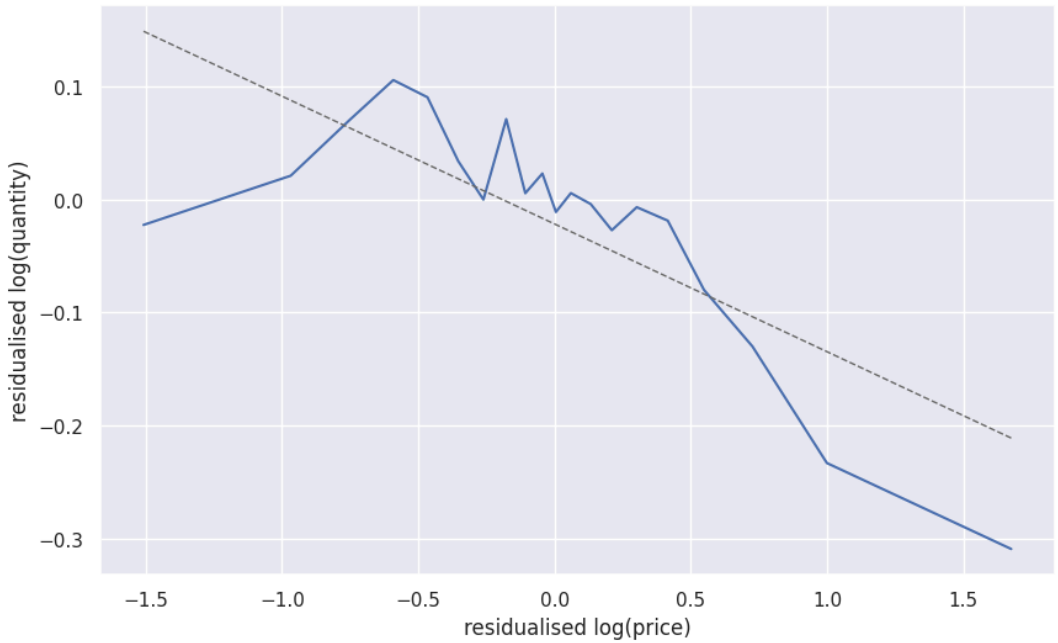


Centered logarithmic negative sales rank over centered logarithmic price as binned plot

Baseline DML Model

- Estimate with tabular data as covariates X_{tab} in `DoubleMLPLR` model with `RandomForest` regressors
  - $l_0(X_{\text{tab}}) := \mathbb{E}[\ln(Q)|X_{\text{tab}}]$
    - $R^2_{l_0} = 0.5986$
  - $m_0(X_{\text{tab}}) := \mathbb{E}[\ln(P)|X_{\text{tab}}]$
    - $R^2_{m_0} = 0.1884$

2.5 %	θ	**97.5 %**
-0.132	-0.1098	-0.080



Residualised logarithmic negative sales rank over residualised logarithmic price as binned plot

Deep Learning Models

- The `DoubleMLPLRDeep` was trained using sets of covariates augmented with text, image, or both modalities:

Used Confounders
$X = (X_{txt}, X_{tab})$
$X = (X_{img}, X_{tab})$
$X = (X_{txt}, X_{img}, X_{tab})$

- With each set of covariates X, 5 runs are evaluated.
- The models were trained over 10 epochs with early stopping callback.
- The data was split into 80% training and 20% validation data.

$\Rightarrow$ The aim is to emphasize the benefits of utilizing unstructured data.

(First) Results

Model[1]	Covariates	$R^2_{l_0}$	$R^2_{m_0}$	$\hat{\theta}_0$
`OLS`	$X = X_{tab}$	0.3300	-	-0.0455
`PLR`	$X = X_{tab}$	0.5986	0.1884	-0.1098
`PLRDeep`	$X = (X_{txt}, X_{tab})$	$0.5574_{\pm 0.025}$	$0.6188_{\pm 0.015}$	$-0.2137_{\pm 0.053}$
`PLRDeep`	$X = (X_{img}, X_{tab})$	$0.4518_{\pm 0.029}$	$0.5085_{\pm 0.028}$	$-0.2721_{\pm 0.038}$
`PLRDeep`	$X = (X_{txt}, X_{img}, X_{tab})$	$0.5829_{\pm 0.015}$	$0.6532_{\pm 0.018}$	$-0.2794_{\pm 0.044}$

- Estimates $\hat{\theta}$ become more negative with the inclusion of unstructured data
- The R^2-scores of the nuisance functions improve with the inclusion of unstructured data
- Semi-synthetic data can illustrate the potential of unstructured data (see Appendix)

Future Directions

- Explore settings with multiple treatment variables.
- Investigate heterogeneous treatment effects.
- Apply the DoubleMLDeep framework in Difference-in-Differences (DiD) and Instrumental Variables (IV) settings.
- Conduct extensive simulation studies to test the robustness and accuracy of the DoubleMLDeep framework under various conditions.

$\Rightarrow$ **You can find more information in our paper: Klaassen et al. (2024)**

Thank you for your attention!



University of Hamburg

https://www.bwl.uni-hamburg.de/en/statistik/allgemein.html



Massachusetts Institute of Technology

https://stat.mit.edu/about/



Economic AI

https://economicai.com/



DoubleML

https://docs.doubleml.org/stable/index.html

References

References

Bach, Philipp, Victor Chernozhukov, Malte S Kurz, and Martin Spindler. 2022. "DoubleML-an Object-Oriented Implementation of Double Machine Learning in Python." *Journal of Machine Learning Research* 23: 53–51.

Bach, Philipp, Malte S. Kurz, Victor Chernozhukov, Martin Spindler, and Sven Klaassen. 2024. "DoubleML: An Object-Oriented Implementation of Double Machine Learning in r." *Journal of Statistical Software* 108 (3): 1–56. https://doi.org/10.18637/jss.v108.i03.

Bajari, Patrick, Zhihao Cen, Victor Chernozhukov, Manoj Manukonda, Suhas Vijaykumar, Jin Wang, Ramon Huerta, et al. 2023. "Hedonic Prices and Quality Adjusted Price Indices Powered by AI." https://arxiv.org/abs/2305.00044.

Chan, Alex J., Isabel Chien, Edward T. Moseley, Saad Salman, Sarah Kaminer Bourland, Daniela Lamas, Anne M. Walling, and James A. Tulsky. 2018. "Deep Learning Algorithms to Identify Documentation of Serious Illness Conversations During Intensive Care Unit Admissions." *Palliative Medicine*. https://doi.org/10.1177/0269216318810421.

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. "Double/Debiased Machine Learning for Treatment and Structural Parameters." *The Econometrics Journal* 21 (1): C1–68. https://onlinelibrary.wiley.com/doi/abs/10.1111/ectj.12097.

Egami, Naoki, Christian J. Fong, Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. 2018. "How to Make Causal Inferences Using Texts." https://arxiv.org/abs/1802.02163.

Jerzak, Connor T., Fredrik Johansson, and Adel Daoud. 2023a. "Estimating Causal Effects Under Image Confounding Bias with an Application to Poverty in Africa." https://arxiv.org/abs/2206.06410.

———. 2023b. "Image-Based Treatment Effect Heterogeneity." https://arxiv.org/abs/2206.06417.

———. 2023c. "Integrating Earth Observation Data into Causal Inference: Challenges and Opportunities." https://arxiv.org/abs/2301.12985.

Ke, Guolin, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. "Lightgbm: A Highly Efficient Gradient Boosting Decision Tree." *Advances in Neural Information Processing Systems* 30: 3146–54.

Klaassen, Sven, Jan Teichert-Kluge, Philipp Bach, Victor Chernozhukov, Martin Spindler, and Suhas Vijaykumar. 2024. "DoubleMLDeep: Estimation of Causal Effects with Multimodal Data." https://arxiv.org/abs/2402.01785.

Krizhevsky, Alex. 2009. “Learning Multiple Layers of Features from Tiny Images.”

Maas, Andrew L., Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. 2011. “Learning Word Vectors for Sentiment Analysis.” In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, 142–50. Portland, Oregon, USA: Association for Computational Linguistics. http://www.aclweb.org/anthology/P11-1015.

Masukawa, Kento, Maho Aoyama, Shinichiroh Yokota, Junya Nakamura, Ryoka Ishida, and Masaharu Nakayama. 2022. “Machine Learning Models to Detect Social Distress, Spiritual Pain, and Severe Physical Psychological Symptoms in Terminally Ill Patients with Cancer from Unstructured Text Data in Electronic Medical Records.” *Palliative Medicine*. https://doi.org/10.1177/02692163221105595.

Melnychuk, Valentyn, Dennis Frauen, and Stefan Feuerriegel. 2022. “Causal Transformer for Estimating Counterfactual Outcomes.” https://arxiv.org/abs/2204.07258.

Sridhar, Dhanya, and David M. Blei. 2022. “Causal Inference from Text: A Commentary.” *Science Advances* 8 (42): eade6585. https://doi.org/10.1126/sciadv.ade6585.

Veitch, Victor, Dhanya Sridhar, and David Blei. 2020. “Adapting Text Embeddings for Causal Inference.” In *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*, edited by Jonas Peters and David Sontag, 124:919–28. Proceedings of Machine Learning Research. PMLR. https://proceedings.mlr.press/v124/veitch20a.html.

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. https://ggplot2.tidyverse.org.

Wu, Zhenhuan, Xingde Duan, and Wenzhuan Zhang. 2023. “Bayesian Analysis of Tweedie Compound Poisson Partial Linear Mixed Models with Nonignorable Missing Response and Covariates.” *Entropy*. https://doi.org/10.3390/e25030506.

Appendix: Double Machine Learning

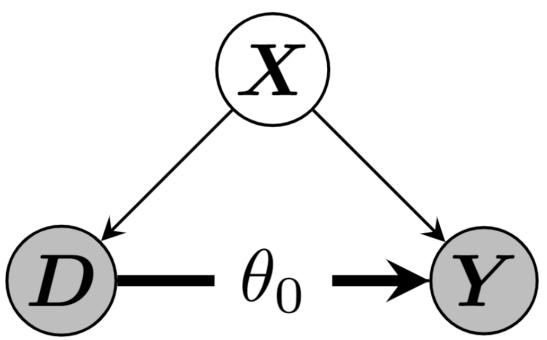
Motivating Example - PLR

- Revisit the PLR model from the beginning:

$$
\begin{aligned}
Y &= \theta_0 D + g_0(X) + \varepsilon, & \mathbb{E}[\varepsilon | X, D] &= 0 \\
D &= m_0(X) + \vartheta, & \mathbb{E}[\vartheta | X] &= 0
\end{aligned}
$$

with

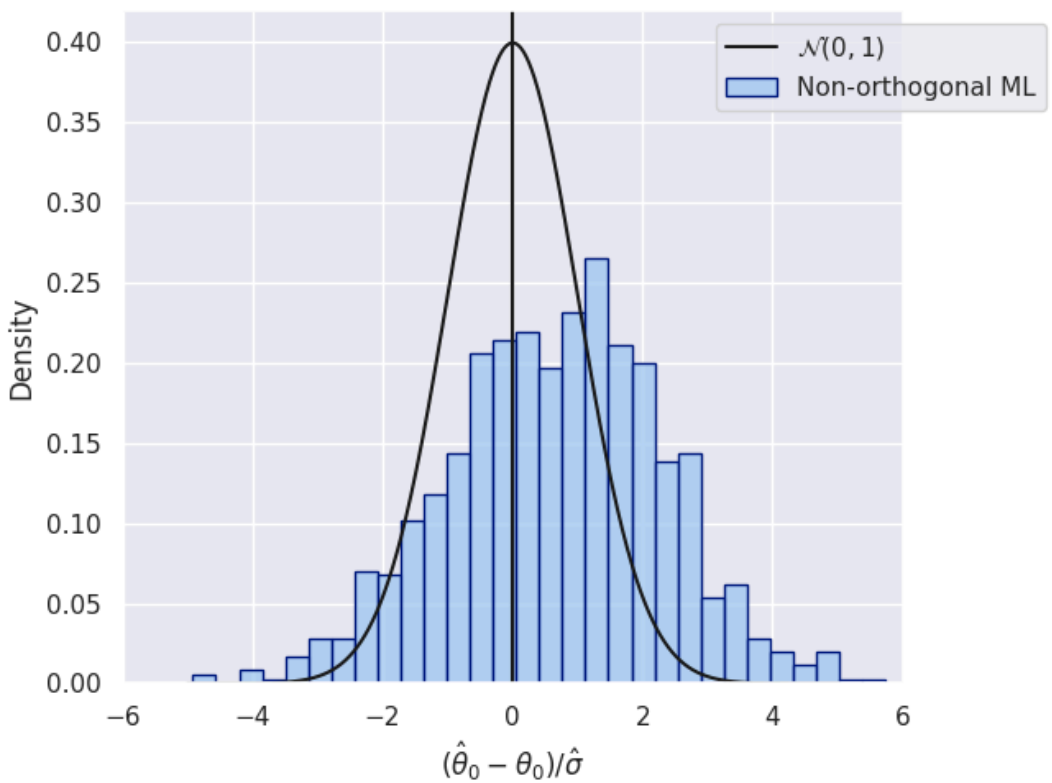
- Y - outcome variable
- D - policy/treatment variable
- $X = (X_1, \ldots X_p)^T$ - vector of additional controls
- ε, ϑ - stochastic errors



DAG for PLR Model

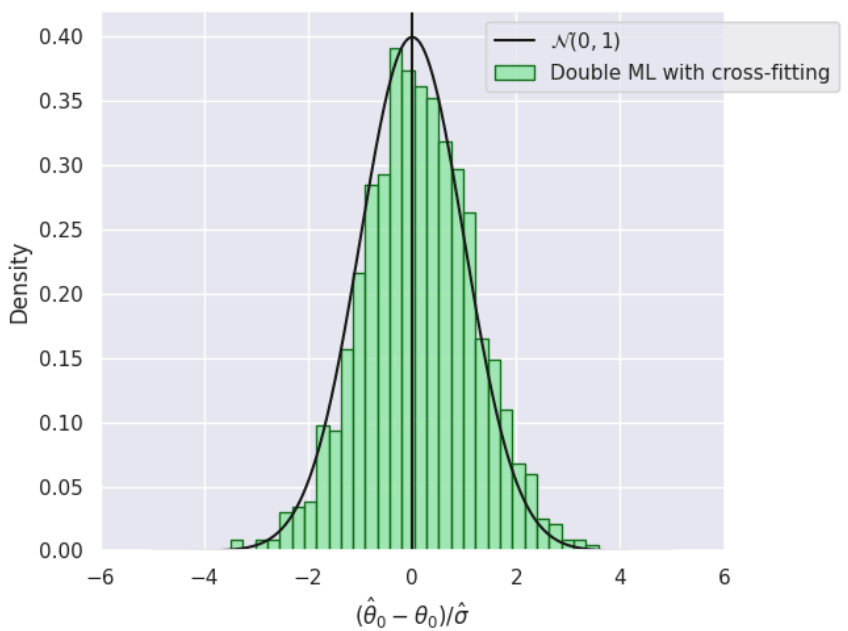
Motivating Example - Regularization Bias

- What if we simply plug-in ML predictions $\hat{g}(X)$ for $g_0(X)$ into $Y = \theta_0 D + g_0(X) + \varepsilon$?



Motivating Example - Orthogonalization

- *Frisch-Waugh-Lovell* style approach: θ_0 can be consistently estimated by **partialling out** X, i.e,
  1. Predict Y and D by $\mathbb{E}[Y|X]$ and $\mathbb{E}[D|X]$, obtained using ML methods
  2. Residualize $\tilde{Y} = Y - \mathbb{E}[Y|X]$ and $\tilde{D} = D - \mathbb{E}[D|X]$
  3. Regress $\tilde{Y}$ on $\tilde{D}$ to obtain $\hat{\theta}$



Double/Debiased Machine Learning (DML)

- **Neyman Orthogonality**

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]|_{\eta=\eta_0} = 0$$

ensures that the moment condition identifying θ_0 is **insensitive to small pertubations** of the nuisance function η around η_0

- Assumptions for the nuisance elements (see Chernozhukov et al. (2018))

$$\|\hat{m} - m_0\|_{P,2} \times \left(\|\hat{m} - m_0\|_{P,2} + \|\hat{\ell} - \ell_0\|_{P,2}\right) \leq \delta_N N^{-1/2}$$

- Under some regularity conditions the estimator $\hat{\theta}$ concentrates in $1/\sqrt{N}$-neighborhood of θ_0 and

$$\sqrt{N}(\hat{\theta} - \theta_0) \to \mathcal{N}(0, \sigma^2)$$

$\Rightarrow$ simple approach to control for confounders, *if they can be used for ML models!*

Neyman Orthogonality

The naive approach minimizes the following MSE

$$min_\theta \mathbb{E}[(Y - D\theta_0 - g_0(X))^2]$$

This implies the following moment equation

$$\mathbb{E}[\underbrace{(Y - D\theta_0 - g_0(X))D}_{=:\psi(W,\theta_0,\eta)}] = 0$$

Whereas for the partialling-out approach minimizes

$$min_\theta \mathbb{E}\left[\left(Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta\right)^2\right]$$

which implies

$$\mathbb{E}\Big[\underbrace{\left(Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta\right)(D - \mathbb{E}[D|X])}_{=:\psi(W,\theta_0,\eta)}\Big] = 0$$

Neyman Orthogonality

Naive approach

$$\psi(W, \theta_0, \eta) = (Y - D\theta_0 - g_0(X))D$$

Regression adjustment score

$$\begin{aligned} \eta &= g(X), \\ \eta_0 &= g_0(X). \end{aligned}$$

FWL partialling out

$$\begin{aligned} \psi(W, \theta_0, \eta_0) = &\Big((Y - E[Y|X]) \\ &- (D - E[D|X])\theta_0\Big) \\ &(D - E[D|X]) \end{aligned}$$

Neyman-orthogonal score (Frisch-Waugh-Lovell)

$$\begin{aligned} \eta &= (\ell(X), m(X)), \\ \eta_0 &= (\ell_0(X), m_0(X)), \\ &= (\mathbb{E}[Y \mid X], \mathbb{E}[D \mid X]). \end{aligned}$$

DML Key Ingredients

1. Neyman Orthogonality

- Inference is based on a moment condition that satisfies the **Neyman orthogonality condition** $\psi(W; \theta, \eta)$

$$E[\psi(W; \theta_0, \eta_0)] = 0,$$

- where $W := (Y, D, X, Z)$ and with θ_0 being the unique solution that obeys the **Neyman orthogonality condition**

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]|_{\eta=\eta_0} = 0.$$

- ∂_η denotes the pathwise (Gateaux) derivative operator

DML Key Ingredients

1. Neyman Orthogonality

- **Neyman orthogonality** ensures that the **moment condition** identifying θ_0 is **insensitive to small pertubations** of the nuisance function η around η_0
- Using a Neyman-orthogonal score **eliminates the first order biases** arising from the replacement of η_0 with a ML estimator $\hat{\eta}_0$
- PLR example: Partialling-out score function

$$\psi(\cdot) = (Y - E[Y|X] - \theta(D - E[D|X]))(D - E[D|X])$$

DML Key Ingredients

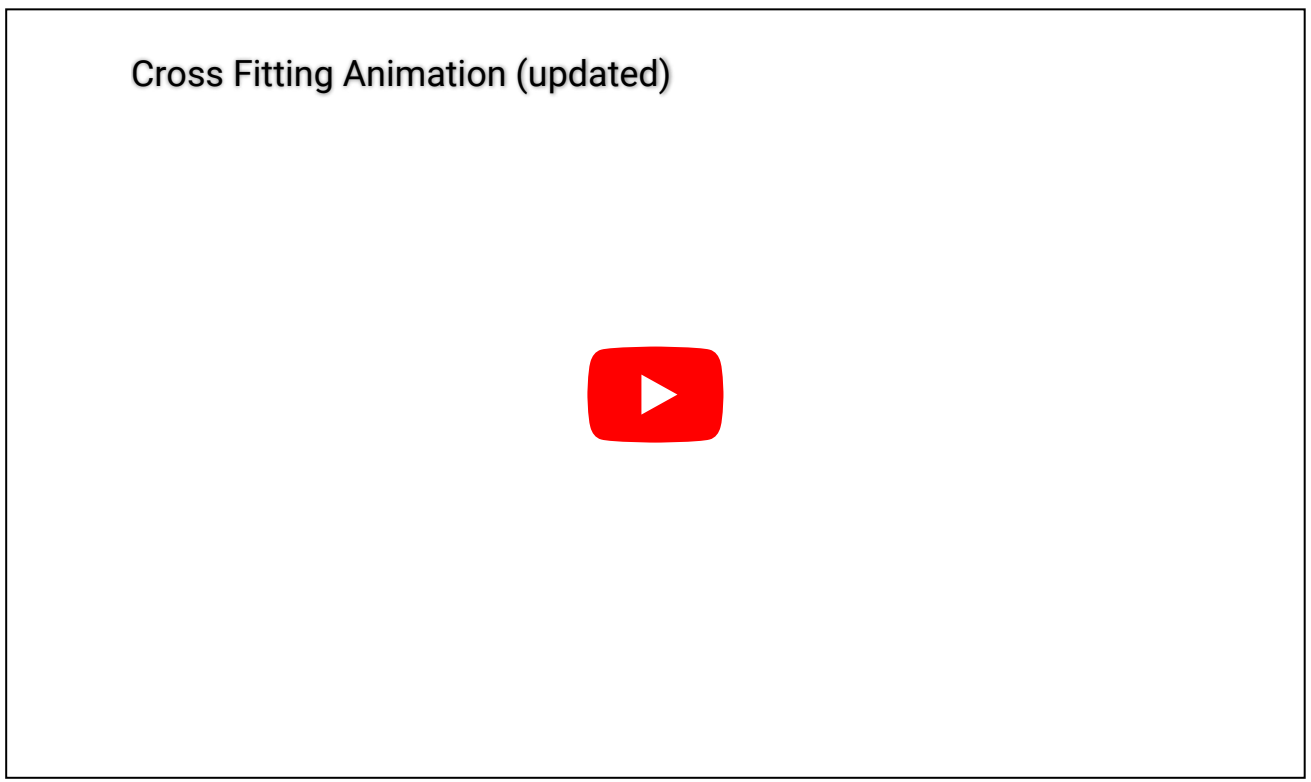
2. High-Quality Machine Learning Estimators

- The nuisance parameters are estimated with high-quality (fast-enough converging) machine learning methods.
- Different structural assumptions on η_0 lead to the use of different machine-learning tools for estimating η_0 Chernozhukov et al. (2018) (Section 3)
- Rate requirements depend on the causal model and orthogonal score, e.g. (see Chernozhukov et al. (2018)),
  - PLR, partialling out:
    $\|\hat{m}_0 - m_0\|_{P,2} \times \left(\|\hat{m}_0 - m_0\|_{P,2} + \|\hat{\ell}_0 - \ell_0\|_{P,2}\right) \leq \delta_N N^{-1/2}$
  - IRM/DR score, ATE: $\|\hat{m}_0 - m_0\|_{P,2} \times \|\hat{\ell}_0 - \ell_0\|_{P,2} \leq \delta_N N^{-1/2}$

DML Key Ingredients

3. Sample Splitting

- To avoid the biases arising from overfitting, a form of **sample splitting** is used at the stage of producing the estimator of the main parameter θ_0.
- Efficiency gains by using cross-fitting (swapping roles of samples for train / hold-out)



DML Key Ingredients

Main result in Chernozhukov et al. (2018)

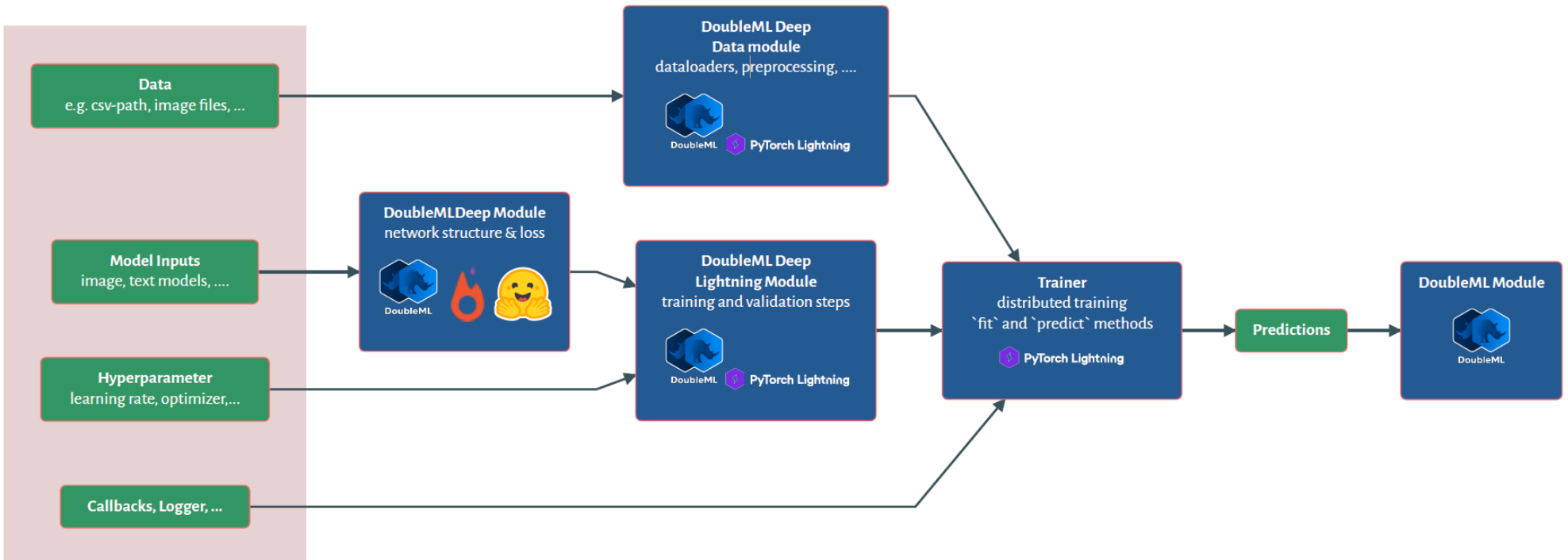
> There exist regularity conditions, such that the DML estimator $\tilde{\theta}_0$ concentrates in a $1/\sqrt{N}$-neighborhood of θ_0 and the sampling error is approximately
>
> $$\sqrt{N}(\tilde{\theta}_0 - \theta_0) \sim N(0, \sigma^2),$$
>
> with
>
> $$\sigma^2 := J_0^{-2}\mathbb{E}(\psi^2(W; \theta_0, \eta_0)),$$
> $$J_0 = \mathbb{E}(\psi_a(W; \eta_0)).$$

- See this example based on Chernozhukov et al. (2018)

DoubleMLDeep - More Details

- Use pretrained models for unstructured data e.g. Bert and Beit
- Basic models for tabular data seem sufficient
- Monitor the nuisance losses
  - $\|Y - \hat{l}(X)\|_{P,2} \leq \|l_0(X) - \hat{l}(X)\|_{P,2} + \sigma_\epsilon$
  - $\|D - \hat{m}(X)\|_{P,2} \leq \|m_0(X) - \hat{m}(X)\|_{P,2} + \sigma_\vartheta$

DoubleMLDeep - Workflow



High-Level Workflow of *DoubleMLDeep*

Appendix: Application (Amazon Toys Data)

Amazon Toys Dataset

Continous Variables

- Reviews: Rating
- Reviews: Review Count
- New Offer Count: Current
- Count of retrieved live offers: New, FBA
- Count of retrieved live offers: New, FBM

Categorical Variables

- Lightning Deals: Upcoming Deal
- Buy Box: Is FBA

Deep Learning Models

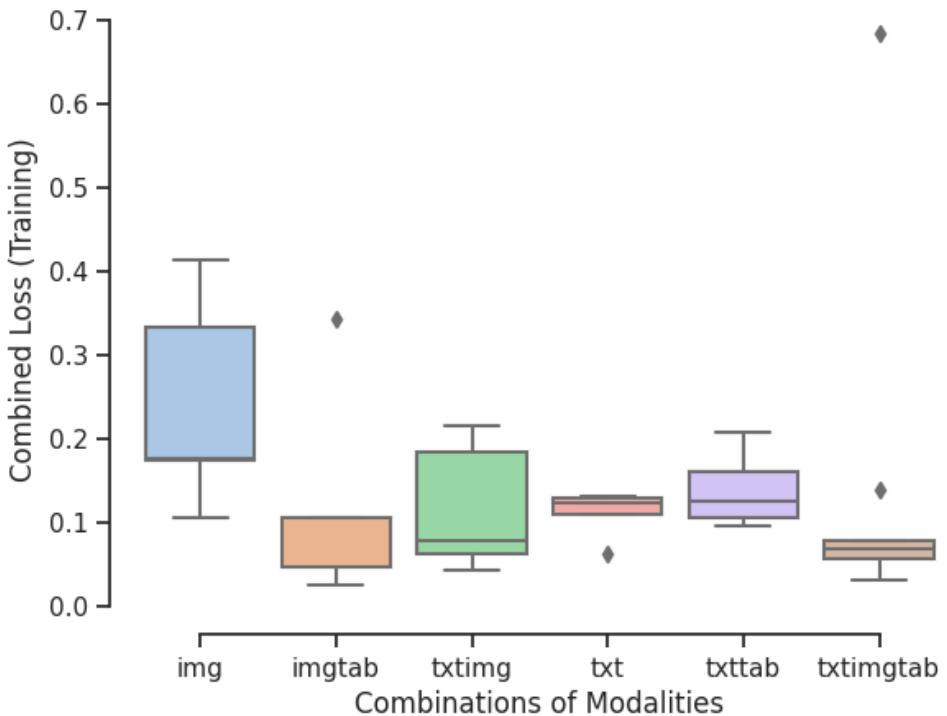
- In this study, a variety of input data combinations were used to train `DoubleMLPLRDeep` models [1]:

Key	Used Confounders
`img`	$X = (X_{img})$
`imgtab`	$X = (X_{img}, X_{tab})$
`txtimg`	$X = (X_{txt}, X_{img})$
`txt`	$X = (X_{txt})$
`txttab`	$X = (X_{txt}, X_{tab})$
`txtimgtab`	$X = (X_{txt}, X_{img}, X_{tab})$

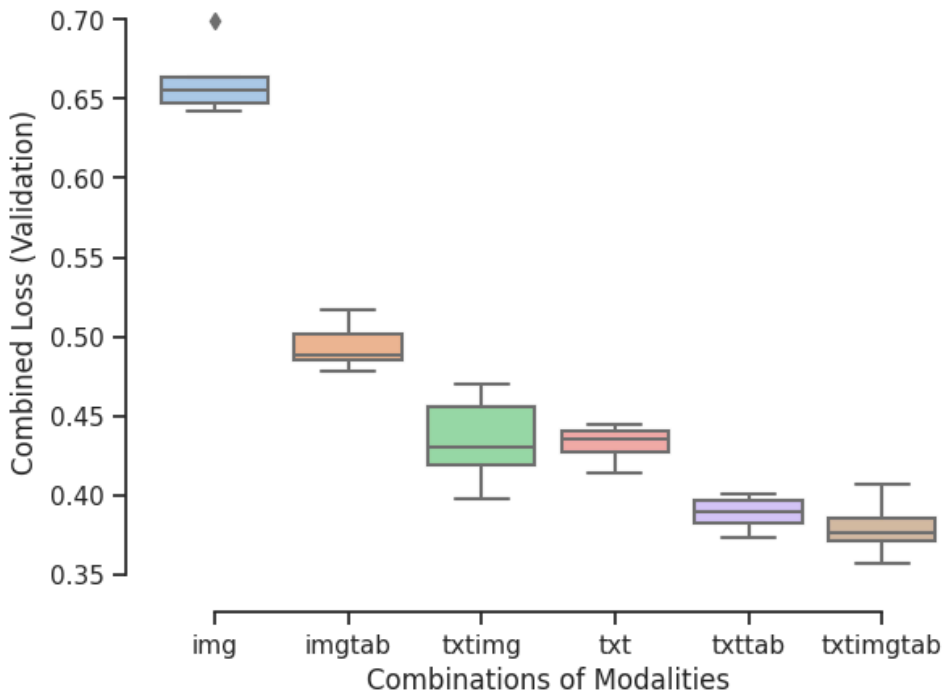
$\Rightarrow$ The aim is to emphasize the benefits of utilizing unstructured data.

Deep Learning Models: RMSE-Scores

- Combined RMSE Score (Training)

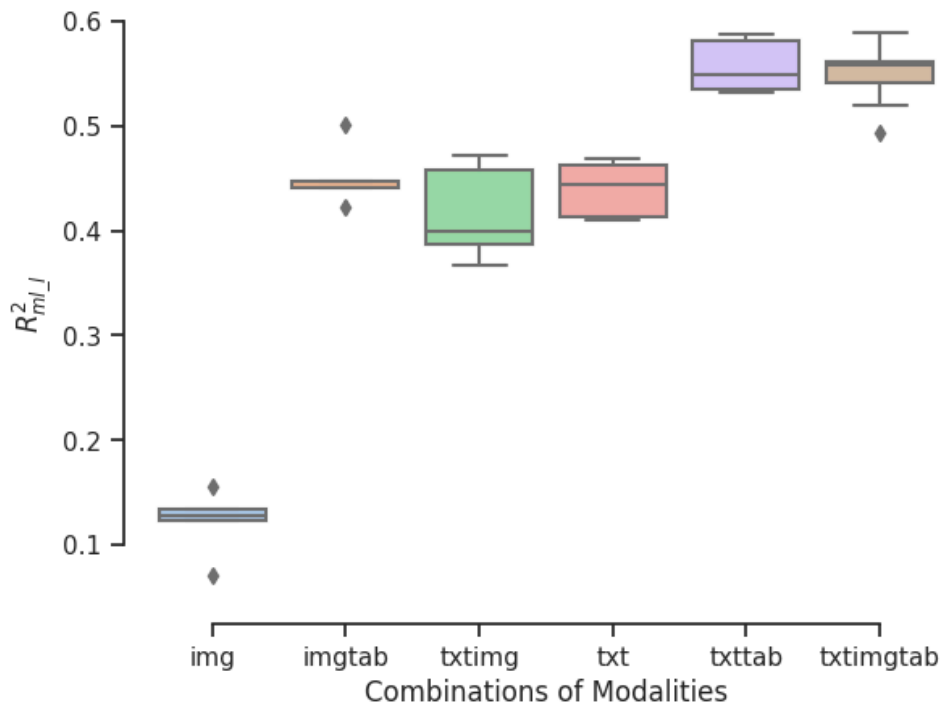


- Combined RMSE Score (Validation)

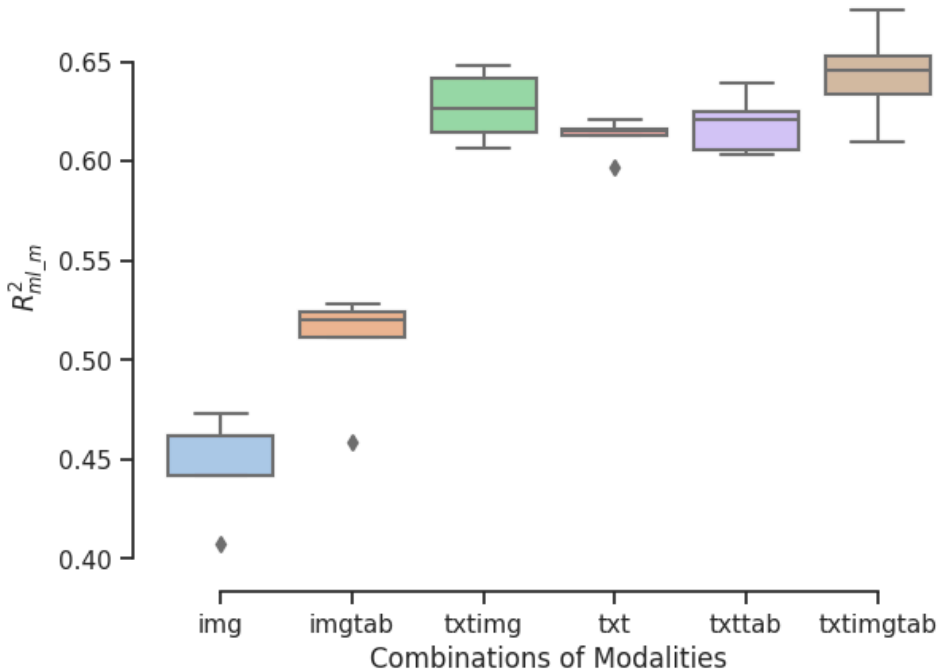


Deep Learning Models: R^2-Scores

- R^2 of log(Quantity) on Validation Set

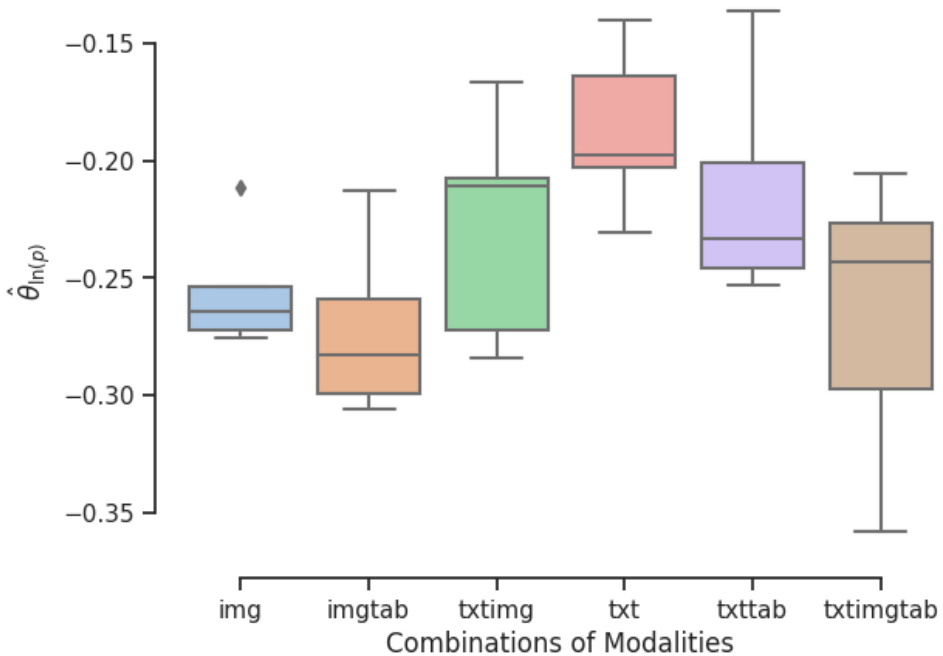


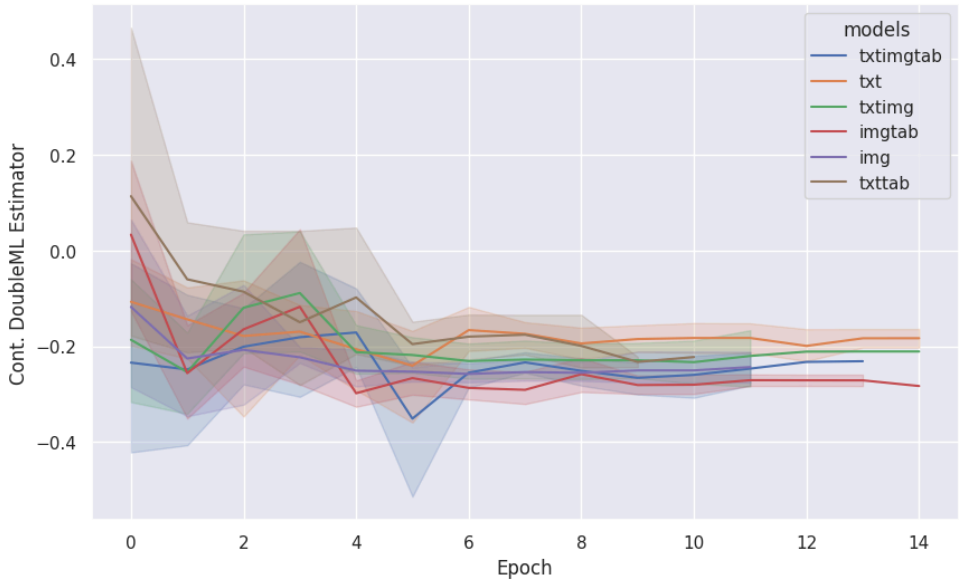
- R^2 of log(Price) on Validation Set



(First) Results

- Using different combinations of tabular, image and text data we obtain the following estimates





Continous Estimates from `DoubleMLPLR` on Validation Set during the Training Process

Appendix: Semi-Synthetic Data

Original Datasets

- We generate a semi-synthetic dataset with a known treatment effect parameter
- To generate credible confounding wil be based on the **labels** or **outcomes** of the corresponding supervised learning task
- We use the following datasets:

Modality	**Dataset**	**Target** $\tilde{X}$	**Control** X
Tabular	DIAMONDS Wickham (2016)	$\log(\text{Price})$	Carat, Cut, Color, Clarity, ...
Text	IMDB Maas et al. (2011)	Sentiment	Review Text
Image	CIFAR-10 Krizhevsky (2009)	Label	Image

Data Generating Process

We generate a semi-synthetic dataset according to the underlying PLR model

$$
\begin{aligned}
Y &= \theta_0 D + \tilde{g}_0(\tilde{X}) + \varepsilon, \\
D &= \tilde{m}_0(\tilde{X}) + \vartheta,
\end{aligned}
$$

where $\tilde{X} = (\tilde{X}_{\text{tab}}, \tilde{X}_{\text{txt}}, \tilde{X}_{\text{img}})$ with the following additive structure

$$
\begin{aligned}
\tilde{g}_0(\tilde{X}) &= \sum_{\text{mod} \in \{\text{tab,txt,img}\}} \tilde{g}_{\text{mod}}(\tilde{X}_{\text{mod}}) \\
\tilde{m}_0(\tilde{X}) &= \sum_{\text{mod} \in \{\text{tab,txt,img}\}} \tilde{m}_{\text{mod}}(\tilde{X}_{\text{mod}})
\end{aligned}
$$

and $\varepsilon, \vartheta \sim \mathcal{N}(0, 1)$.

Data Generating Process

- The effect on the outcome Y is generated via a standardized version of target variable to balance the confounding impact of all modalities:

$$\tilde{g}_{\text{mod}}(\tilde{X}_{\text{mod}}) = \frac{\tilde{X}_{\text{mod}} - \mathbb{E}[\tilde{X}_{\text{mod}}]}{\sigma_{\tilde{X}_{\text{mod}}}}, \quad \text{mod} \in \{\text{tab}, \text{txt}, \text{img}\}$$

- Further, to ensure a strong confounding, the impact on the treatment D is defined via:

$$\tilde{m}_{\text{mod}}(\tilde{X}_{\text{mod}}) = -\tilde{g}_{\text{mod}}(\tilde{X}_{\text{mod}}), \quad \text{mod} \in \{\text{tab}, \text{txt}, \text{img}\}$$

- The treatment effect is set to $\theta_0 = 0.5$ with $n = 50.000$ samples in the dataset
- Both $\tilde{g}_0(X)$ and $\tilde{m}_0(X)$ are rescaled to ensure a signal-to-noise ratio of 2 for Y and D (given unit variances of the error terms)

Challenges

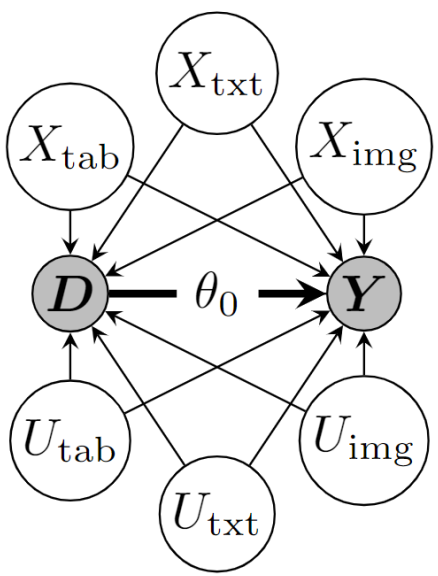
- Dependency on the target of the modality $\tilde{X}_{\text{mod}}$
  - Might not be fully explained by the corresponding features X_{mod}
  - E.g. price of diamonds can not be fully explained by the carat, cut, color, etc.
- Consequently, the estimate $\hat{\theta}$ might only be able to account for the part of confounding which can be explained by the input features:

$$\tilde{X}_{\text{mod}} = \mathbb{E}[\tilde{X}_{\text{mod}} | X_{\text{mod}}] + U_{\text{mod}},$$

  where U_{mod} can not be controlled for

Challenges

- Due to the negative sign and the additive structure, the confounding effect will ensure that higher outcomes Y occur with lower treatment values D, creating a **negative bias**
- The independence of all three original datasets and the additive negative confounding results in a negative bias even if we only control for a subset of confounding factors



DAG for the semi-synthetic dataset. The confounding via the features $X = (X_{\text{tab}}, X_{\text{txt}}, X_{\text{img}})$ can be adjusted for, whereas the unexplained/noise parts $U = (U_{\text{tab}}, U_{\text{txt}}, U_{\text{img}})$ are unobserved.

Bounds for the estimation of θ_0

- Since all modalities contribute a negative bias, the semi-synthetic dataset can be used as a benchmark with an oracle upper bound of an effect estimate of

$$\theta_0 = 0.5$$

- For the lower bound, one can not account for confounding at all and evaluate a basic ordinary least squares model with outcome Y on the treatment variable D (excluding all confounding variables)

$$\hat{\theta}_{\text{OLS}} = -0.4594$$

Bounds for the estimation of η_0

- To access the predictive performance of the nuisance models, we can rely on oracle predictions of

$$\begin{aligned}\tilde{m}_0(\tilde{X}) &:= \mathbb{E}[D|\tilde{X}] \\ \tilde{l}_0(\tilde{X}) &:= \mathbb{E}[Y|\tilde{X}] = \theta_0 \tilde{m}_0(\tilde{X}) + \tilde{g}_0(\tilde{X})\end{aligned}$$

- Evaluating the oracle predictions $\tilde{m}_0(\tilde{X})$ and $\tilde{l}_0(\tilde{X})$ results in the following upper bounds for the performance of the nuisance estimators

$$\begin{aligned}R^2(D, \tilde{m}_0(\tilde{X})) &= 0.6713 \\ R^2(Y, \tilde{l}_0(\tilde{X})) &= 0.5845\end{aligned}$$

  on the whole dataset of $n = 50.000$ observations

Bounds for the estimation of η_0

- In order to compare the predictive performance of different models, a relative r^2-score with respect to the upper bound is defined as

$$0 \leq r^2(D, \hat{m}) := \frac{R^2(D, \hat{m}(X))}{R^2(D, \tilde{m}_0(\tilde{X}))} \leq 1$$

$$0 \leq r^2(Y, \hat{l}) := \frac{R^2(Y, \hat{l}(X))}{R^2(Y, \tilde{l}_0(\tilde{X}))} \leq 1$$

Models

- **Baseline Model**
  - Standard DML approach, relying only on tabular data X_{tab}
  - Based on the LightGBM package for estimation of the nuisance elements
- **Deep Model**
  - Uses the out-of-sample predictions of $\hat{m}(X)$ and $\hat{l}(X)$ generated from the model
  - Based on the RoBERTa Model for text modality, a VIT Model for images and a SAINT model for the tabular data
- **Embedding Model**
  - Generated embedding H_E is used together with the tabular features X_{tab} as input for LightGBM
  - Based on the RoBERTa Model for text modality, a VIT Model for images and a SAINT model for the tabular data

Simulation Results

Results of Simulation Study. Reported: mean ± sd. over five random train-test splits

	Baseline	**Embedding**	**Deep**
$r^2(Y, \hat{l}_0)$	0.31 ± 0.01	0.87 ± 0.02	$\mathbf{0.90 \pm 0.01}$
$r^2(D, \hat{m}_0)$	0.31 ± 0.01	0.87 ± 0.02	$\mathbf{0.90 \pm 0.01}$
$\hat{\theta}$	-0.32 ± 0.01	$\mathbf{0.28 \pm 0.01}$	0.27 ± 0.01

Higher = better (best in bold)

Simulation Results - Performance of $\hat{\theta}$



Boxplots of $\hat{\theta}$. The Embedding Model and Deep Model have DoubleML Deep estimates. This indicates a stable and information-rich